Linear Algebra I – Week 6
Independent Study: 40 Proposed Problems

Tashkent University of Technology

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Reading: G. Strang, *Introduction to Linear Algebra*, 4th ed., Sections 2.4–2.7.
Instructions: Show elimination steps, verify matrix products, and justify when an inverse or factorization exists. Problems progress from routine to challenge.

Part 1 – 2 by 2 foundations

1. 2 by 2: inverse. Find A^{-1} for $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}$. Verify by multiplication.

2. 2 by 2: solve. Solve $Ax = b$ with $A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$.

3. 2 by 2: inverse. Find A^{-1} for $A = \begin{bmatrix} 4 & 2 \\ 1 & 4 \end{bmatrix}$. Verify by multiplication.

4. 2 by 2: solve. Solve $Ax = b$ with $A = \begin{bmatrix} 2 & 1 \\ 2 & 5 \end{bmatrix}$ and $b = \begin{bmatrix} 4 \\ 12 \end{bmatrix}$.

5. 2 by 2: factorization. Find $A = LU = LDU$ for $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$.

6. 2 by 2: factorization. Find $A = LU = LDU$ for $A = \begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix}$.

7. 2 by 2: factorization. Find $A = LU = LDU$ for $A = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$.

8. 2 by 2: factorization. Find $A = LU = LDU$ for $A = \begin{bmatrix} 2 & 1 \\ 4 & 4 \end{bmatrix}$.

Part 2 – 3 by 3 elimination

9. 3 by 3: elimination. Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$.

10. 3 by 3: elimination. Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix}$.

11. 3 by 3: elimination. Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 6 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}$.

12. 3 by 3: elimination. Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$.

13. 3 by 3: LU and LDU. Compute LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$.

14. 3 by 3: LU and LDU. Compute LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 4 & 2 \\ -2 & 5 & 3 \end{bmatrix}$.

15. 3 by 3: LU and LDU. Compute LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 6 & 3 \\ -2 & 11 & 5 \end{bmatrix}$.

16. 3 by 3: LU and LDU. Compute LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$.

Part 3 – Triangular factorizations

17. Row exchange and PLU. Find a row permutation P and factors satisfying $PA = LU$ for $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$.

18. Transpose and inverse. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$, compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

19. Symmetric LDL transpose. Factor symmetric $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ as LDL^T.

20. Challenge: tridiagonal model. Solve $Ax = b$ by LU for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest-neighbor structure.

21. 2 by 2: inverse. Find A^{-1} for $A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$. Verify by multiplication.

22. 2 by 2: solve. Solve $Ax = b$ with $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $b = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$.

23. 2 by 2: inverse. Find A^{-1} for $A = \begin{bmatrix} 4 & 1 \\ 2 & 5 \end{bmatrix}$. Verify by multiplication.

24. 2 by 2: solve. Solve $Ax = b$ with $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}$ and $b = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$.

Part 4 – Permutations and transposes

25. 2 by 2: factorization. Find $A = LU = LDU$ for $A = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$.

26. 2 by 2: factorization. Find $A = LU = LDU$ for $A = \begin{bmatrix} 3 & 1 \\ 3 & 5 \end{bmatrix}$.

27. 2 by 2: factorization. Find $A = LU = LDU$ for $A = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$.

28. 2 by 2: factorization. Find $A = LU = LDU$ for $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$.

29. 3 by 3: elimination. Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$.

30. 3 by 3: elimination. Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$.

31. 3 by 3: elimination. Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}$.

32. 3 by 3: elimination. Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$.

Part 5 – Applications and challenges

33. 3 by 3: LU and LDU. Compute LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$.

34. 3 by 3: LU and LDU. Compute LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$.

35. 3 by 3: LU and LDU. Compute LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$.

36. 3 by 3: LU and LDU. Compute LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$.

37. Row exchange and PLU. Find a row permutation P and factors satisfying $PA = LU$ for $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$.

38. Transpose and inverse. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$, compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

39. Symmetric LDL transpose. Factor symmetric $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ as LDL^T.

40. Challenge: tridiagonal model. Solve $Ax = b$ by LU for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$.

Interpret the nearest-neighbor structure.